BIRD HOUSES
BATHS AND FEEDING SHELTERS

HOW TO MAKE AND WHERE TO PLACE THEM

EDMUND J. SAWYER

CRANBROOK INSTITUTE OF SCIENCE

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EDMUND J. SAWYER

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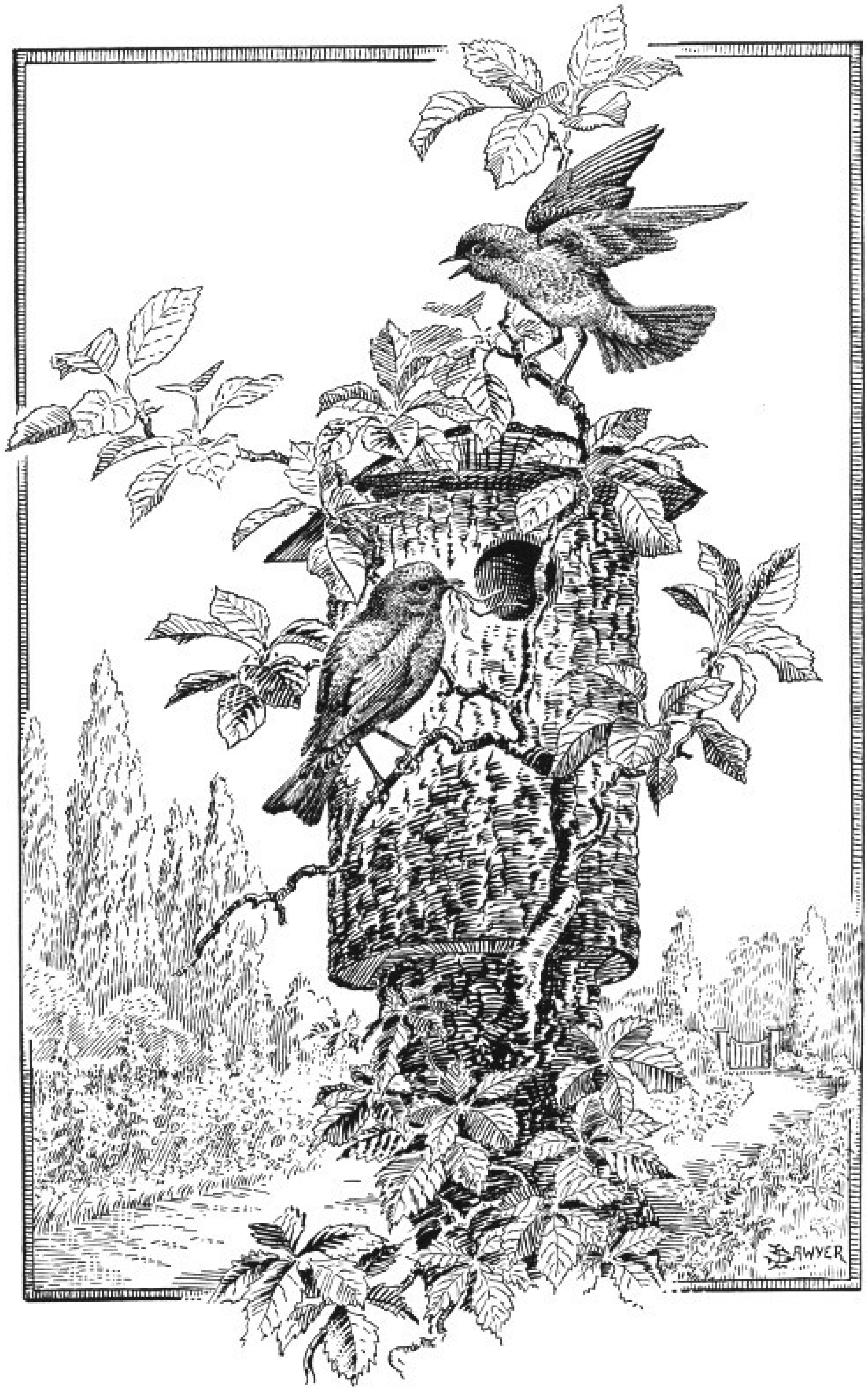

Bluebirds

Foreword

Most species of the smaller birds which nest in hollow trees, and therefore in bird houses, suffer seriously from intrusion by English Sparrows and European Starlings. These two species, nesting in similar locations and being prolific, tend to take up all available nesting cavities, even ejecting native birds which have built or begun to build. This condition, already serious, may become far more baneful than we are as yet able to realize. It may even contribute to the eventual extinction of Bluebirds, Crested Flycatchers, and Purple Martins unless we provide nesting sites sufficient in number and suitable in kind for all. The number of natural nesting sites is already far below that required by these birds, and yet the Starlings in particular are increasing alarmingly. There is no way to determine when a final adjustment or balance will be reached or what the numerical status of our native bird-house dwellers will be when such balance shall have been attained.

In the case of the native species named above, we can at least help by providing proper nesting boxes which will induce the birds to concentrate about our houses, where the European Starling and the English Sparrow menace may be more easily and effectively met than elsewhere. The reader should note that the imported English Sparrow, which is in reality a weaver finch and unrelated to our tuneful native sparrows, is the only "sparrow" east of the Rocky Mountains that nests in holes or bird houses. The shyer, more desirable, native species are invariably harmless and should be both protected and encouraged.

The smaller of the bird-house species, such as the House Wren, Tree Swallow, Black-capped Chickadee, White-breasted Nuthatch, and Tufted Titmouse are less affected by the intruders. These smaller birds can use, and usually select, cavities with openings too small for either Starling or English Sparrow to enter. However, there is still a distinct practical advantage in providing proper boxes for House Wrens and other small species. In many instances these birds fail to find places safe from the ubiquitous English Sparrow and European Starling, which then proceed in their well-known manner to work destruction. A properly made wren house, chickadee house, or swallow house can be absolutely safe from the foreign invaders.

More nesting sites, properly made and situated, are desirable, owing to 6
the destruction of birds' traditional nesting sites over wide acreages about cities and other settled areas, but the providing of suitable bird houses needs no defense or excuse. Whether it be the beautiful and demure bluebird, "bird of happiness," the sleek and immaculate swallow, the songful wren, "saucy and impudent," the bustling and industrious chickadee, or the alert and noisy flycatcher, the native tenant of the bird house will be an interesting and entertaining neighbor, always prompt to pay his rent in one form or another or in many forms and with interest. Does one need any special excuse for offering hospitality to such a neighbor? When birds nest on our home grounds, their destruction of garden pests, mosquitos, gnats, and other undesirable insects is concentrated where we can most directly profit from the results—our own greater comfort and safety, better gardens, more productive orchards, more verdure in shade trees and in ornamental trees and shrubs.

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Bird Houses and Common Sense

Although there are a number of points which should be considered in the proper designing and placing of a bird house, there is one simple idea which practically covers the whole subject. Every species of our small native birds that nests in a bird house nested originally in a hollow tree, by preference in a hollow of one unvarying type—the burrow made by a woodpecker. Thus we need only know what the burrow of a woodpecker is like and we have automatically solved in a general way the questions of material; size and shape of entrance; diameter, depth and form of cavity; height above ground; and situation. The nature of nesting material and its whereabouts should play absolutely no part in human plans for the prospective tenants. “Unfurnished” rooms are the only kind for which birds are looking.

There is solid ground for assuming a woodpecker’s burrow to be the ideal pattern for a bird house. The woodpecker, whatever its species, free to excavate any form of chamber that it might wish, invariably uses *one* type of burrow. The birds which by preference habitually adopt for their own use the woodpecker’s abandoned home have likewise thus placed their own age-old stamp of approval on that type. It is logical to assume, therefore, that the artificial bird house should follow at least the general plan of that long-tried and preeminently successful nesting site. Since a theory may be plausible while yet utterly untenable in actual application, it remains to add that abundance of experience in building and placing bird houses all goes to prove the foregoing basic principle soundly correct in practice. With or without benefit of the plans and specifications in such a bulletin as this, a person who takes his cue from a woodpecker will not go far wrong. In planning a bird house, we must continually hark back to the idea of the woodpecker's burrow—or rather, we should never quite lose sight of it.

Some Current Notions Corrected

Attention should be called to some common misconceptions. The colony bird house, or any bird house with more than one compartment, is always a mistake unless it has been designed for Purple Martins. Yet certain 8
firms have for years been advertising "wren houses" of four or more chambers. One who knows this pugnacious little bird tries in vain to imagine two pairs of wrens living peaceably under one small roof! Every bird house should consist of one, and only one, chamber—with the single exception of a house intended for the Purple Martin, which nests in colonies.

The cubic capacity of the bird houses one sees is nearly always much too great—often several times too great. Builders seem to believe that the diameter of the nesting chamber should at least equal the total length of the bird—a theory as erroneous as it is plausible. Plate I illustrates the fact that the sitting bird normally occupies a space measuring much less from side to side than the outstretched length of the bird. Figure No. 4, Plate I, shows how much work is often made for the House Wren, while figure No. 5, on the same plate, shows how greatly this work may be reduced—with the greater inducement to the prospective occupant.

The square or rectangular door is another frequent mistake—a projection of the designer's own plantigrade and vertical personality.

To place the entrance at or near floor level is also an error. Remember that birds close no doors against drafts, that their "beds" are laid on the floor and consist of light straws, feathers, or other flimsy materials.

Many a wren house with entrance (as it should be) too small for any English Sparrow to enter, is hung *swinging from a branch* as a further protection against the unwanted sparrow. That is like beheading a criminal and then, “just to be on the safe side,” shooting him into the bargain! It is said that wrens do not hesitate to use these swinging nesting sites, but we have our serious doubts. We have personally seen one instance of a wren nesting in the pensile home of a Baltimore Oriole, but it is significant that in this case we failed to find any better site nearby. Some persons report success with this type of house and prefer it because of the ease of putting it up and taking it down without injury to a living tree.

Two doors, presumably entrance and exit, to a bird house of one compartment is nearly as ridiculous an innovation as the two doors said to have been provided by a famous scientist for the use of his old cat and her kittens, respectively.

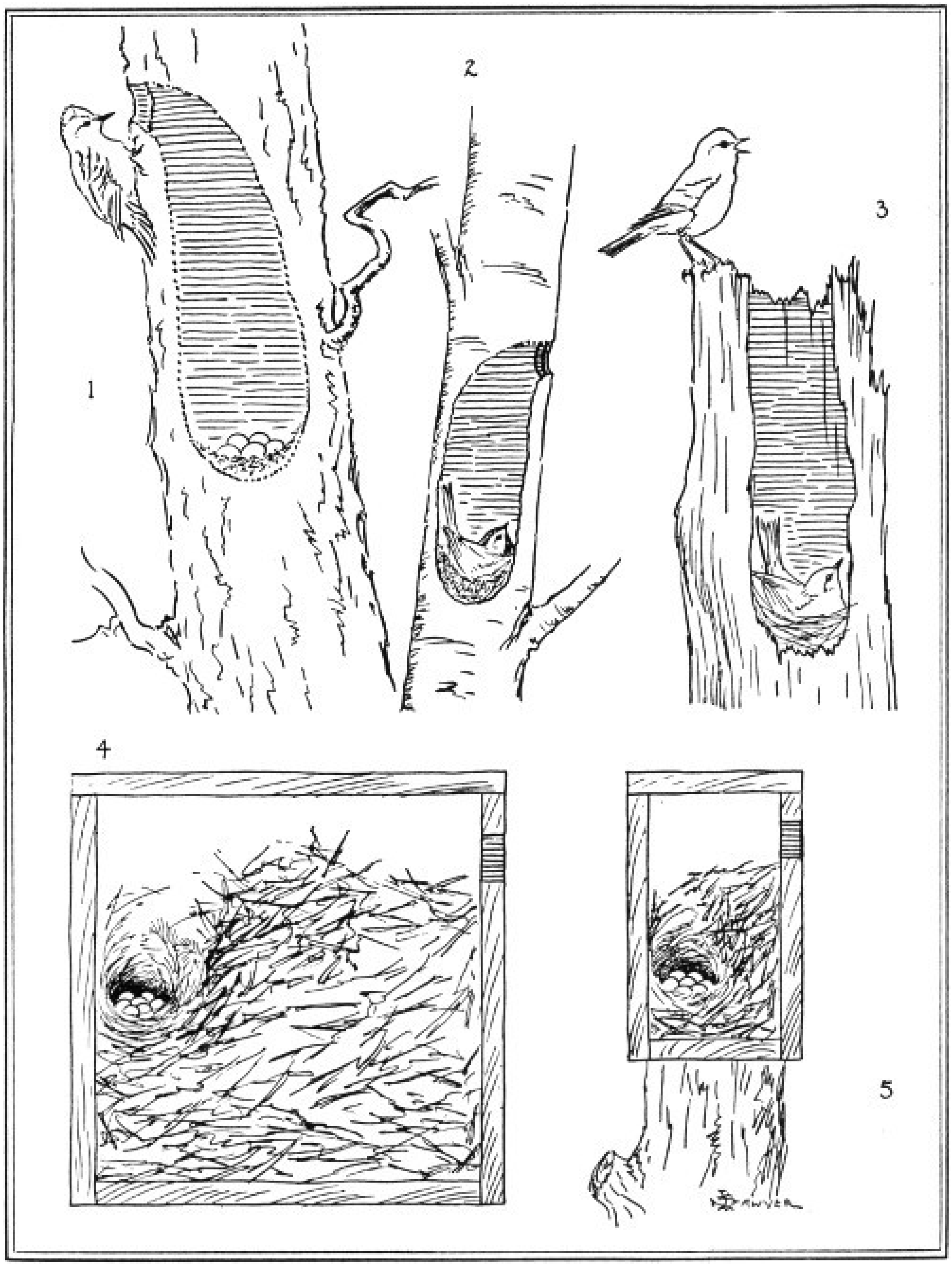


PLATE I. Nesting Sites, Natural and Artificial

1, 2, 3, A nesting woodpecker, a Chickadee, and a pair of bluebirds, respectively. Compare length of bird with diameter of nesting chamber. 4, A wren house, as frequently made, of eight to ten times the necessary cubic capacity. 5, A wren house of proper and ample size.

Overcrowding is a prevalent fault. On an area insufficient properly to harbor two pairs of wrens or bluebirds there will often be a half-dozen or more bird houses. Tree Swallows are social birds and will occupy boxes placed near to one another, but ordinarily, birds, especially those of the same species, do not build near each other. It is a large town lot which will properly accommodate more than one pair of nesting wrens. Even the demure bluebirds do not like to build within a stone’s throw of each other. While the martin colony may number upward of a dozen pairs in the same house, there may not be other martins within a mile. There is many a small village whose single martin house accommodates all the martins to be found within a radius of several miles. 10

The size of entrance seems often to be a stumbling block. One sees wren houses with perfect bluebird entrances, and bluebird houses with doorways best suited to wrens or chickadees or, at the other extreme, to doves!

Although arguments, supported by some experience, have been advanced for larger entrances, we nevertheless suggest entrances of nearly minimum size—a suggestion based on personal experience and long familiarity with the preferences shown by the species concerned. Apparently John Burroughs was first to point out that when birds hesitate to enter a small opening it is evidently because their bodies, completely filling the entrance, render the cavity totally dark and therefore alarming. Cut a few small auger holes to admit light, and the bird enters the now somewhat less mysterious chamber. The holes also provide needed ventilation, but they should be small and well above the entrance-level, for drafts must be avoided. The entrance to the house for wren, chickadee, or Tree Swallow should be, since it easily may be, too small to admit English Sparrows. It is not possible to exclude English Sparrows from houses of other birds in that way.

Finally, the mistake is often made of providing simply *a* bird house instead of a *martin* house, a *wren* house, a *bluebird* house, or a house for some other definite species. The result is that such houses usually go unoccupied or else are promptly claimed by the first English Sparrows that spy them. Any bird house will suit the English Sparrow if only he can get into it, and he usually can get into *a* bird house. So avoid type *a*—the too common variety.

General and Miscellaneous

Material

Wood is the material par excellence for the bird house, the only material which can be unreservedly recommended. Substitutes have been used with varying degrees of success or failure. What glass is for the window, wood is for the bird house. First, the birds are habituated to it. It is a good nonconductor of heat, it resists rain and extremes of temperature, and it can be made to harmonize with its setting. Over a long period of time it improves, rather than suffers, from exposure to the weather.

Soft wood with straight grain, such as pine or spruce, is preferable. It is easily worked, may be nailed with little danger of serious splitting, and is sufficiently durable. Slab wood, with or without the bark, and old fence-boards make the most generally effective bird houses. If new lumber is used, it should be rough, not planed; whether rough or planed, it should be treated with gray, olive, or dull brown stain of a medium shade. There are numbers of suitable oil stains on the market. The stain should permeate the grain of the wood, without actually coating it as paint does. Thickness of the boards should not be far from 1 inch. If slab wood is used, it may be anywhere from ⅜ inch at the edge, up to about 2 inches in the thickest part.

Leave the matter of materials for the nest itself entirely to the birds. Not only is it quite unnecessary to place twigs, straws, strings, or even choice feathers or other fluffy bits on or near the bird house, but such action tends actually to defeat its own purpose. The prospective tenants often seem to regard these well-meant efforts as evidence of a competitor who has the advantage of priority. This is especially apt to be the case when the materials are placed inside the house. Birds will sometimes steal nesting sites and even the raw materials of others but may choose to avoid the clash which such piracy entails, provided there are other sites and materials not too far to seek.

Entrance

There is always danger of the wood splitting when too large an auger hole is attempted. Before assembling the bird house, make an entrance in the front board. Start by drawing a circle the exact size of the doorway-to-be. Then, just inside this circle, bore four holes at equal intervals, using a bit not larger than ⅜ inch for the smaller entrances; not larger than ¾ inch for the larger entrances. It will now be not too difficult, by use of a keyhole saw or pocketknife and wood-rasp, to remove the wood still remaining inside the drawn circle. Placing the board horizontally in a vice will further insure against splitting while the holes are being bored.

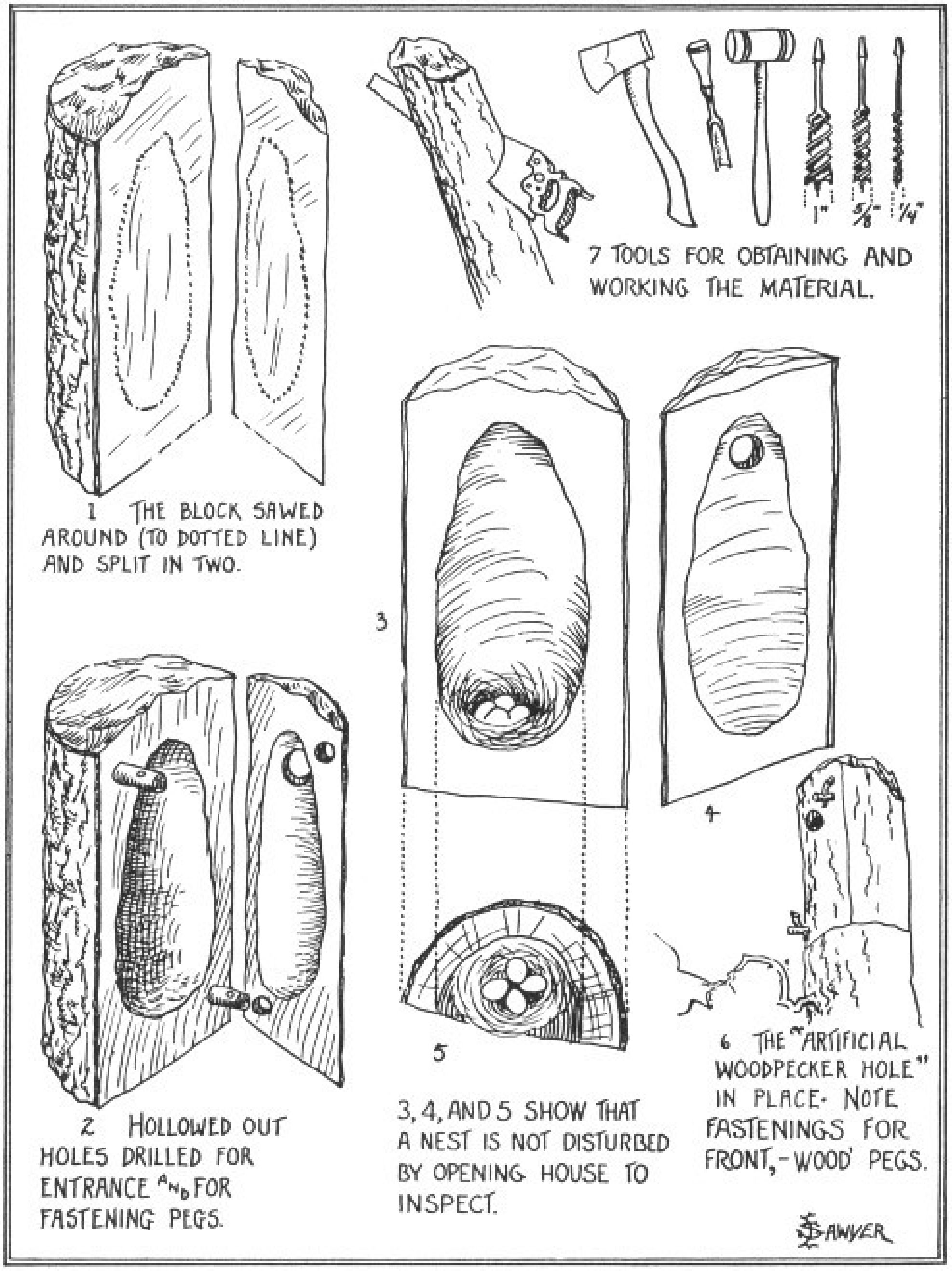


PLATE II. The Best and Most Simple Form of Artificial Nesting Site

With dimensions appropriate to the species, this is the most generally attractive type of house for all species excepting the Purple Martin. It is the type especially suited for the chickadee, nuthatch, and Tufted Titmouse.

1 THE BLOCK SAWED AROUND (TO DOTTED LINE) AND SPLIT IN TWO.
2 HOLLOWED OUT HOLES DRILLED FOR ENTRANCE AND FOR FASTENING PEGS.
3, 4, AND 5 SHOW THAT A NEST IS NOT DISTURBED BY OPENING HOUSE TO INSPECT.
6 THE "ARTIFICIAL WOODPECKER HOLE" IN PLACE. NOTE FASTENINGS FOR FRONT,—WOOD PEGS.
7 TOOLS FOR OBTAINING AND WORKING THE MATERIAL.

Nails

When a bird house is fastened to a support by the "toe-nail" method (by driving nails at a slant through the sides and bottom), it is a good point first to drill holes of the right diameter to fit the nails tightly; otherwise splitting of the sides or bottom of the house may result. Use flat-headed nails.

Slabs

When slabs are used in house-building, the upright pieces may be fastened to each other, at intervals of several inches, with a wire staple having ¾-inch prongs one inch apart. These should be on the outside of the house, where rust will color them to conform with the rustic wood.

Facilities for House Cleaning

For inspecting the nest and, at the end of the season, for cleaning out the old nest material, the top or some other section of the house should always be easily removable. Exceptions to this rule are houses for ducks and other larger birds, for which the entrance may be large enough and the depth not too great for all such purposes. Do not open a house in the owner's presence. The more brief and infrequent your inspections, the less they will disturb the birds.

Position of Boxes

Bird houses erected on poles are safer from predators than those placed in trees. Houses for Purple Martins, in particular, need to be at a distance from trees and buildings, and if possible near water.

Place your bird house where the sun will reach it during part of the day, and turn the entrance away from the prevailing winds.

It seems hardly necessary to emphasize that, if possible, the bird 14
house, as well as the bird bath and feeding station, should be placed in full view of a convenient window. To watch birds in their building and other activities will prove a fascinating pleasure.

Undesired Tenants

The author once with complete success contrived and operated a mechanical “bouncer” to meet a particular and aggravated instance of bird trespassers. In case of interference with any desired tenant or prospective tenant by rivals for the same bird house, the interfering birds may be driven away by this device.

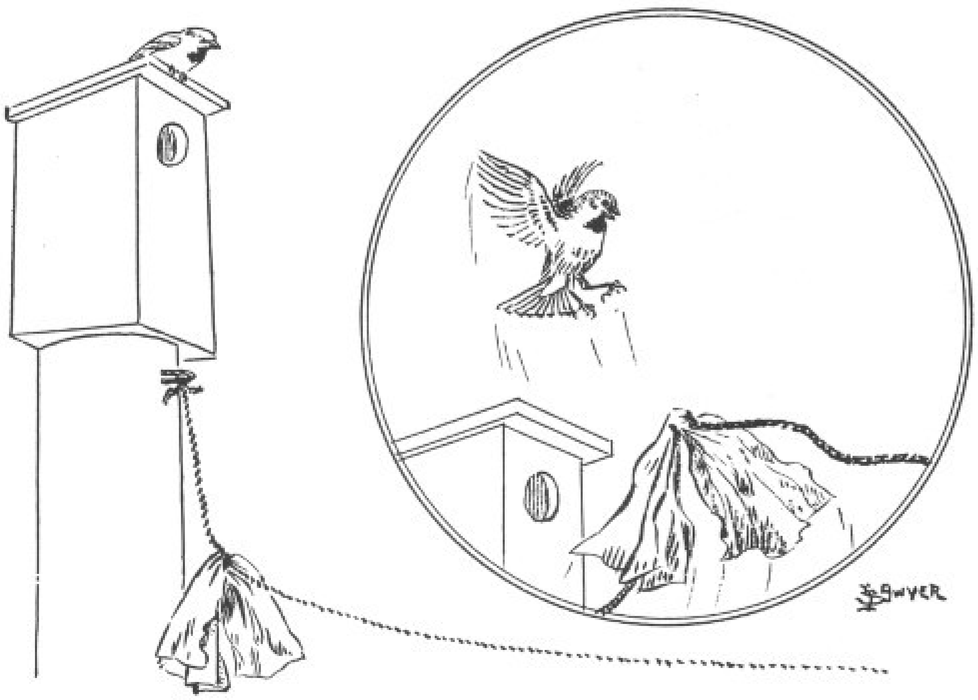


FIGURE 1. Discouraging the Uninvited Guest.

One end of a stout cord is attached at some point close below the bird house. To this cord a rag or a piece of waxed paper about man's-handkerchief size is tied as conspicuously as possible at a distance from the fastened end of the cord about equal to the height of the bird house. The cord's free end is then carried through a window of your own house from which there is a convenient view of the bird box. The end of the cord is fastened inside the window where it can be easily reached and jerked, 15
enough slack being allowed to let the rag at the other end hang down.
If the unwanted birds appear on or too close to the bird house, the cord is given a sharp pull which will cause the rag to jerk upward and frighten the intruder away. Repeat as often as occasion requires.

Care should be taken to work the bouncer when only the undesired birds are within effective distance. However, should the contending birds actually come to bodily grips on or beside the nesting site, the fight may be broken up by vigorous operation of the alarm. The author found that this device, exercised as occasion required, in two days ended a week-old continuous struggle between a Bluebird and a pair of Tree Swallows in which the Bluebird had entirely defeated the daily efforts of the Swallows to start a nest. The Swallows were left in possession, quickly built their nest, and duly raised their brood.

Obviously, this device and procedure may be applied against Starlings, English Sparrows, or other birds, regardless of species, which for any reason one desires to discourage from building in a given bird house or other nesting-site.

The use of a rag or similar object, instead of a bell or other noisemaker, is advised because the range of alarm is thereby limited to the immediate location of the bird house, so that the desired tenants will not be frightened by the operation of the alarm when they are within hearing.

Thickets

“Reserves” of the favorite environment are as much a need for some species as bird houses are for others. Widespread “improvement” and “beautification” along roadsides is destroying the thicket, the favored haunt of Song Sparrows, Catbirds, certain warblers, quail, and other desirable species. A “reserve” thicket may be located in a secluded part of the home grounds, hidden by a hedge if considered unsightly. It may be almost any shape, but not less than 20 feet in average diameter or much less than 400 square feet in area. Here weeds, tall grass, briars, and dense bushes are allowed to grow wild, forming a tangle as they will. If the bushes include such kinds as hackberry, hawthorn, sumac, elderberry, and chokecherry, the thicket’s annual period of usefulness will be extended. Including such a winter food tree as the mountain ash will invite Purple Finches, Pine Siskins, grosbeaks, waxwings, and perhaps crossbills.

Dimensions for Various Houses

Following are specifications and remarks on the housing requirements of the birds by species. For related forms not included in the table see text.

Table I

	Diameter of interior (inches)	Depth from entrance (inches)	Diameter of entrance (inches)	Distance from ground to entrance (feet)
House Wren	4¼-5½	7-9	1	8-18
Black-capped Chickadee	3¼-4	7-9	1⅛	8-15
White-breasted Nuthatch, Tufted Titmouse	3¾-4½	8-10	1½	12-25
Tree Swallow	4-5½	6-8	1⅜	8-30
Eastern Bluebird	4-5	8-10	1⅝	8-20
Crested Flycatcher	5½-6½	9-12	2⅛	15-40
Flicker	6½-7½	12-16	2½	10-35
Purple Martin	6-7½	6-8	2⅛-2½	12-14

Wood Duck, Hooded Merganser	7½	12-15	4	10-20
Sparrow Hawk	6½-7	14-16	3	20-50
Saw-whet Owl	6-7½	14-16	2⅞	15-45
Screech Owl	7-8	14-16	3¼	15-30

Nesting Box

Common House Finch	4½-6		Open	10-30

Nesting Shelves

	Width (inches)	Length (inches)	Height from ground (feet)
Robin	5-6	8 or more	8-30
Phoebe	3½-4½	7 or more	8-20

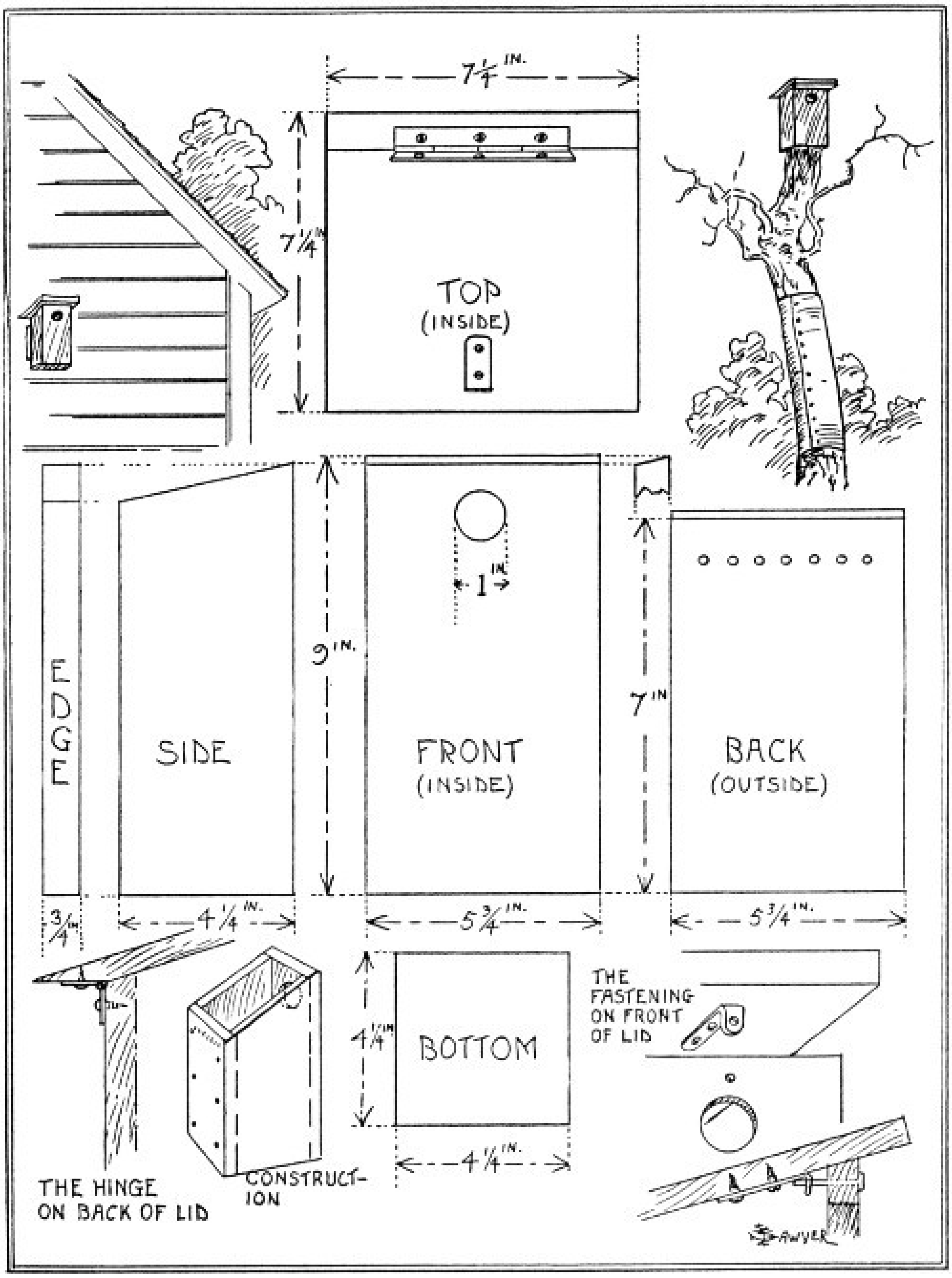


PLATE III. A Simple and Effective Box Bird House

For House Wrens, Bluebirds, and Tree Swallows, especially. For these the house may be of weathered fence-boards or even, if need be, of new lumber stained some dull tint. For chickadees, nuthatches, and Tufted Titmice, rough “slab” material is preferable. Dimensions given are for the House Wren. For other species, see Table I.

TOP (INSIDE): 7¼ IN. × 7¼ IN.

EDGE: ¾ IN. × 9 IN.

SIDE: × 4¼ IN.

FRONT (INSIDE): × 5¾ IN.

{HOLE DIAMETER}: ¾ IN.

BACK (OUTSIDE): 7 IN. × 5¾ IN.

BOTTOM: 4¼ IN. × 4¼ IN.

CONSTRUCTION

THE FASTENING ON FRONT OF LID

THE HINGE ON BACK OF LID

The House Wren

Nesting in all sorts of nooks and crannies, the House Wren is easily satisfied. Moreover, this is often the only desirable species which can be induced to build in so civilized and restricted a place as a small city lot. Slab construction may be employed or old weathered boards used, but new lumber seems nearly or quite as welcome to the wren.

Plate III illustrates a wren house and gives directions for building. Distance from the ground to the entrance of the house should be from 8 to 18 feet. The place safest from cats is on the side of a building. If the box is on a tree or wooden post, protection is afforded by a band of smooth sheet metal such as zinc, 2½ feet high, starting not less than 4 feet from the ground or other point that is within reach of cats.

Two or three wren houses spaced as far apart as the yard or garden allows will provide for the duplicate, unused, nests which these birds often build, or for a possible second brood.

Other Wrens

For the Bewick’s Wren, which commonly nests around gardens, barns, and dwellings, the building directions are the same as for the House Wren.

The Carolina Wren of the south, more inclined to seek woods and thickets than to court man’s society, is not ordinarily a bird house tenant. Still, a home like that described for the House Wren, but with the entrance having a diameter of 1¼ inch, is not unlikely to be selected by the Carolina if placed in a brushy area frequented by him and not much frequented by humans.

The Black-capped Chickadee

Lacking the semi-domestic status of wren and martin, the Black-capped Chickadee is not a regular bird-house addict. He prefers the seclusion of some unfrequented wood. Yet, not uncommonly, he is enticed by a bird house. The specifications and illustrations for the wren house will do for the chickadee. However, a cylindrical and smaller chamber with somewhat larger entrance (1⅛ inch) is more likely to appeal to his uneducated 19
taste. He is still a bird "with the bark on," and his house should be quite literally in keeping. The author personally much prefers to select a hollow branch, from which he cuts a foot-long section. He then drills an entrance hole near one end, nails a piece of slab in place for the bottom, provides a removable lid of the same material, and thus constitutes himself a proxy for the Downy Woodpecker in providing the chickadee with a home. (See Plate II.)

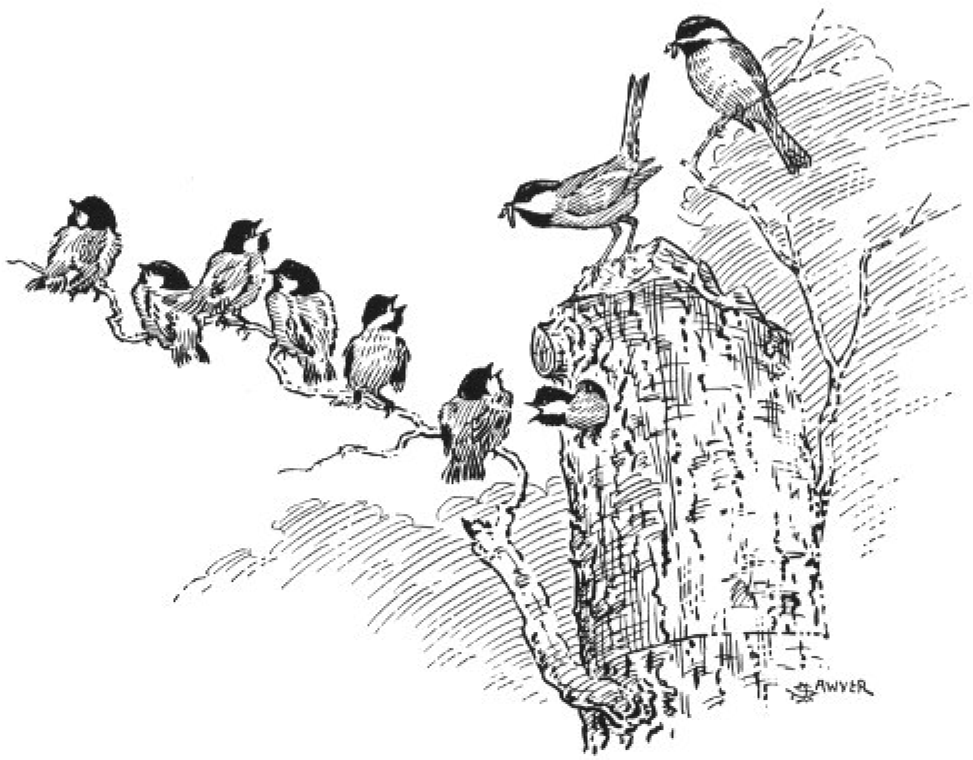


FIGURE 2. A Chickadee Family.

A wood lot or a neglected old orchard bordering a wood is the place for the chickadee house. See Table I for measurements.

Other Chickadees

The Mountain and other chickadees share the natural nesting habits of the Black-cap. Since this group is composed of birds of practically uniform size, a house for any species should be the same as that described for the Black-cap.

However, chickadees are in general birds of the woods. Few or none come so near to being birds of “home grounds” as do the Black-caps. Houses for them, therefore, will be appropriate chiefly about woods homes or camps.

The White-breasted Nuthatch

All that has been written of the chickadee and his housing problem applies also in a general way to the Nuthatch, except that a slightly larger entrance is needed for the latter bird (see measurements in Table I). The Nuthatch, somewhat more than his second cousins, the chickadees, is given to natural knot-holes in living trees. This propensity may well be humored by giving him a house of the hollowed-out trunk variety. If the trunk or branch so hollowed out has a knot which can be made of proper size to serve for the entrance, then that is to add the ultimate artistic finish, the final delicate “touch of nature.” See Plate II.

Other Nuthatches

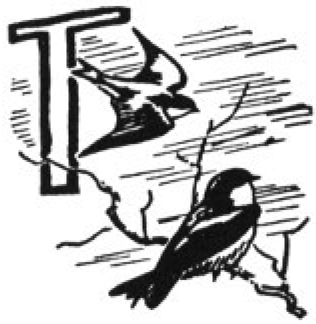
The western subspecies of the White-breasted Nuthatch occur chiefly in less settled or more restricted ranges, and they are apt to be less accustomed to, as well as less accessible for, bird houses.

The Red-breasted, Brown-headed, and other small nuthatches share the hollow-tree nesting habit common to the family. They prefer locations of the wilder kind, so that the usual bird house is not likely to entice them away from less sophisticated haunts. Yet it appears likely that any of the species might select an imitation of its natural nesting site. Build as for the White-breasted, except that the entrance should be but 1¼ inch in diameter.

The Tufted Titmouse

The Tufted Titmouse is common in the south, where its distribution is also much more uniform than in the north. In some places it has pushed far toward our northern border, but there it is inclined to be of only local and irregular occurrence. In general it becomes more and more a bird of the wilder areas as it advances northward. Usually a wood or woodside location will be the one most likely to entice this species to a bird house. When and where the bird is found to linger about residences toward the nesting season, the houses may be placed much as for the bluebird or wren, but in a very quiet corner of the yard. Dimensions are given in Table I (page 16). See also Plates II and III.

The Tree Swallow



The Tree Swallow is often a bird of the small city, but may be expected more dependably in the country or in the rural community. He is easily satisfied as to a nesting place but it is sometimes difficult or impossible for him to find a nook about our dwellings which is safe from the English Sparrow's intrusion. When nesting about inhabited buildings he cannot well afford to dispense with the vigilance of his human landlords. Small, quiet, and peace-loving, he is a particularly poor match for the pugnacious sparrow. Specifications for the Tree Swallow house are given in Table I. See also Plate III. The inner wall, between floor and entrance, must be rough or provided with cleats, in order to give the young a sure foothold. Several pairs of Tree Swallows will willingly nest near one another.

Other Swallows

For the beautiful Violet-green Swallow build exactly as for the Tree Swallow. Where, in the west, both species occur, there is the chance that a house intended for the Violet-green will be taken by the Tree Swallow. This chance, otherwise more than an even one, may be lessened by catering to the Violet-green's observed choice of local haunts. Barn Swallows will sometimes use ledges, such as those described for the Robin (page 29), when these are sheltered.

The Eastern Bluebird

Beautiful, cheerful, demure, raising two or more broods in the season, always a picture of “content in a cottage,” the Bluebird is the all-round ideal tenant for the simple bird house. He is, however, not a bird of the city, nor always a bird of the village garden. The suburbs, especially the more secluded spots therein, suit him much better. He tends, and that with ample reason, to shrink from the society of English Sparrows whose rough, aggressive manners and harsh notes introduce discord into his naturally calm and peaceful existence. By all means provide a house for this 22 peerless tenant; also be prepared to lend him all possible assistance in policing his property until the eggs are laid. Once the precious eggs are deposited, trust the prospective parents to defend their treasure; for even the Bluebird is no exception to the general rule that a brooding bird will readily put an erstwhile successful bully to speedy and inglorious flight. See Plate III and Table I for housing the Bluebird.

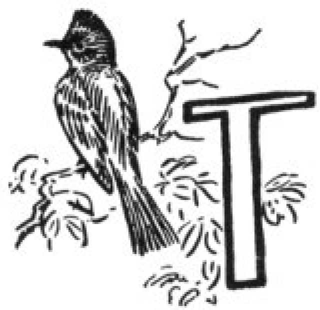
To provide for a second nesting and possibly a third, the procedure should be the same as that described for the House Wren. See, therefore, directions given for that species.

Other Bluebirds

The Western Bluebird fills the same role on the ranges and ranches that is taken on the smaller farms of the east by his eastern namesake. Build and place the house for the one precisely as for the other. The western bird seems much more inclined than the eastern to adopt nesting sites in or close to human dwellings.

The Mountain Bluebird will use the same type of house; its location, of course, should correspond with the local haunts of the species.

The Crested Flycatcher



There is a peculiar satisfaction in successfully providing a house for those species, such as the Crested Flycatcher, which do not ordinarily resort to artificial nesting sites. It is something of a “feather in the cap.” Select a dilapidated orchard or an out-of-the-way woodside as a location for the Flycatcher’s house; place the house and await the results with expectations not too sanguine. Should fortune favor you with an opportunity to watch these birds building, remember to look for the famous dried snake skin almost invariably worked into the nest by this species, not forgetting that the reason for its use is still one of the mysteries. Measurements for the Flycatcher’s house will be found in Table I. See Plates Plate II and Plate III for styles of house recommended.

The Flicker

In general the woodpeckers choose to build their own houses. But as the Flicker is so unlike other woodpeckers in appearance and in certain of his well-known ways, it is not surprising to learn that he will readily take possession of an artificial bird house. Naturally, a woodpecker (and the Flicker is that) is scarcely an exception to the rule that bird-house tenants prefer something along the general lines of a woodpecker's work. A section of a hollow trunk or branch of the proper dimensions inside may easily be transformed into an ideal Flicker house. Next best is the dugout type illustrated in Plate II. Finally, the semi-cylindrical or even rectangular house will do very well if the other specifications are about those given in Table I. The country or the suburbs, not too near to a residence, is the right environment for the Flicker.

A Flicker which took possession of a house I had placed for Crested Flycatchers spent days in audibly widening the rectangular chamber until a soft bed of chips was provided to receive the eggs. The moral is—make the sides of thick, soft wood, and let even a woodpecker furnish his own bedding. However, one or two handfuls of coarse sawdust thrown into the Flicker house will be quite welcome to this bird.

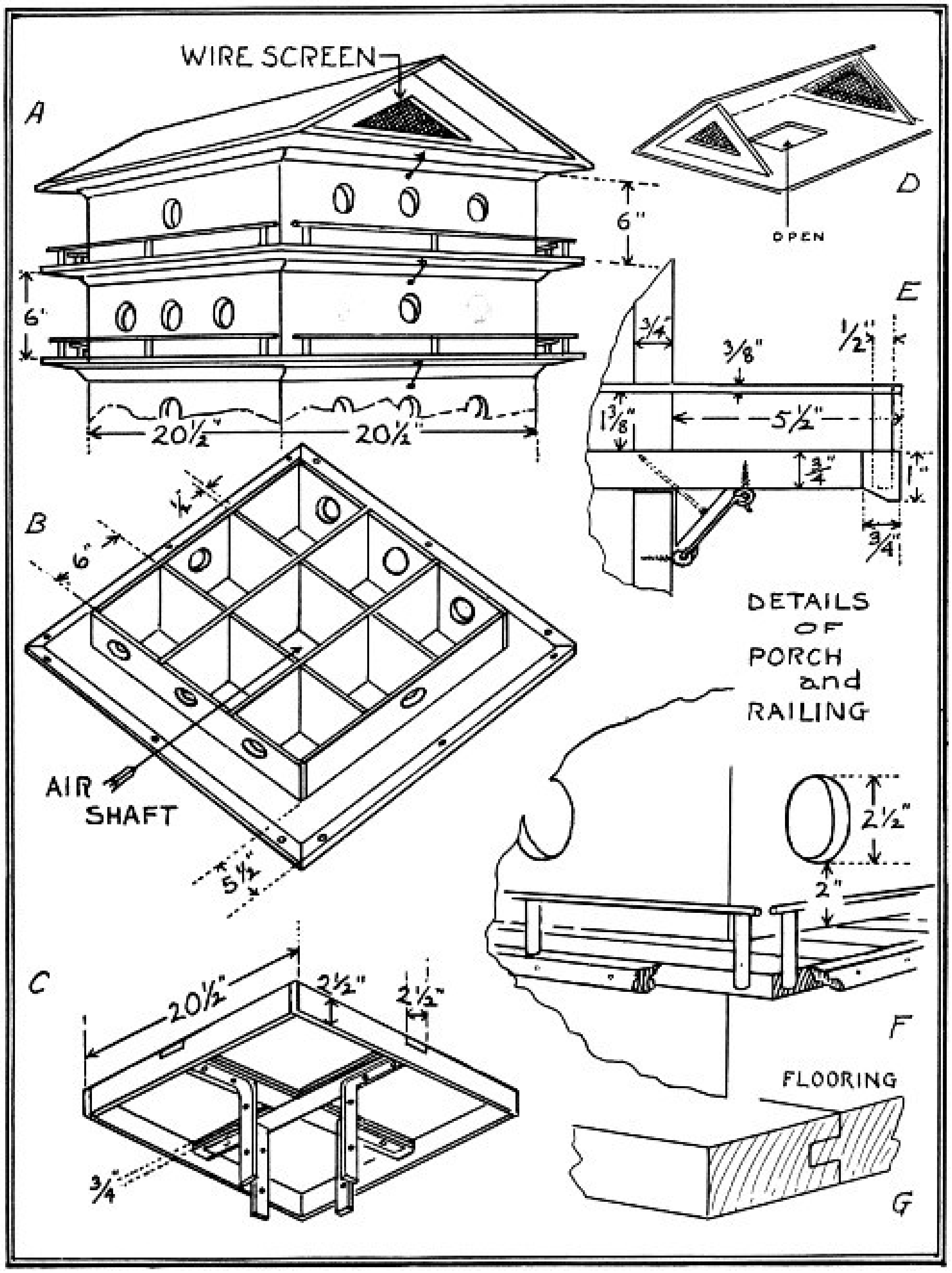
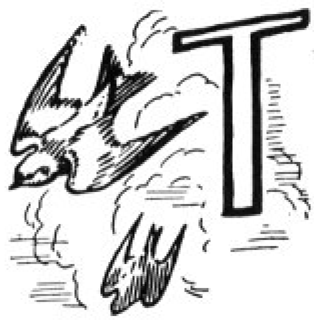


PLATE IV. Martin House

The foundation, each story, and the roof are built as units of uniform lateral dimensions. This permits adding more stories as the colony grows, and allows for easy cleaning. The central air shaft and elevated roof provide cooling by air circulation. A molding attached to the under side of the roof section and to the floor of each other section holds the section in place, aided by hooks and screw eyes as shown. A, Roof and upper stories assembled. B, Interior of one story. The bottom is cut out of the central chamber for the air shaft. C, Foundation or base. Its central cross-pieces are double thicknesses of ¾-inch oak; the rest of the frame is of pine ¾ inch thick. Four heavy angle irons (as shown) attach the base to the pole. D, Interior of roof section exposed to show outlet of air shaft and the screen-covered gable-end air vents (⅛- to ¼-inch mesh screen). E, F, G, Details of porches and railings. The railings and their supporting posts are of standard hardwood dowel stock.

The Purple Martin



This is the only desirable colony-forming bird-house tenant. Therefore the apartment type of house is a waste of material unless intended for Purple Martins and designed accordingly. Of the desirable bird-house clientele, none is quite so sophisticated as the Martins in the matter of a satisfactory location. If it is only so much as a biscuit-toss from the ground, the martin house may grace a bandstand, a village railroad station, or a busy village square. The house itself may be one of those adapted doll houses, complete with chimneys, windows, fancy doorways, and whatnots, ornate in fluted columns, bizarre in lightning rods and weathercocks, pretentious with elaborate porches and other gewgaws, and gaudy with rainbow tints. Go as far as you like, the Martin will pace you. However, for those who would consider the bird's point of view to be of greater importance than their own, appropriate suggestions are offered in Table I and in Plate IV. If painted white the house will be cooler and may be preferred by the birds. The reader should be advised that Martins are temperamental and will sometimes refuse to occupy a suitable house because of some dislike for its situation. Furthermore, Martins sometimes inexplicably abandon a locality where they have previously been abundant.

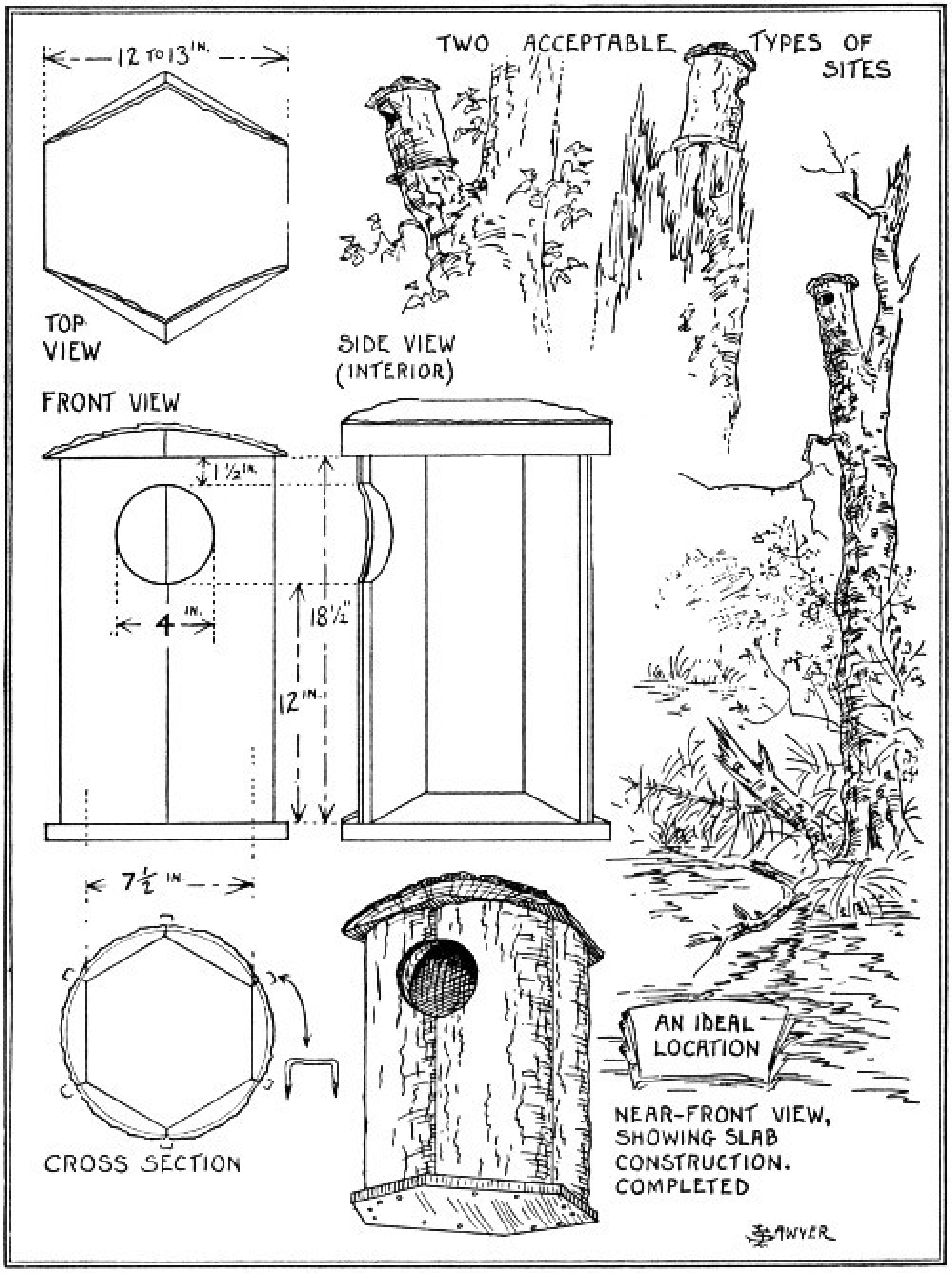
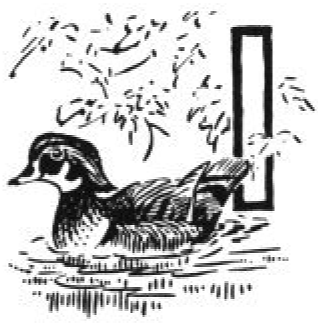


PLATE V. Nesting House for Ducks

The Wood Duck and (in proper places) the Hooded Merganser and Golden-eye are the duck species for which we may provide these houses. The location should be secluded and near the water; the exact site, 10 to 20 feet up on a stub or tree. Build preferably of rough slab material. See text.

The Tree-nesting Ducks



In suitable locations artificial sites may be provided for any of the several wild ducks which ordinarily nest in hollow trees. These ducks, as breeding species, are mostly northern, the Wood Duck being almost the only one which regularly nests, except at the higher elevations, very much south of the northern United States border. The Hooded Merganser may appropriate the house intended for the Wood Duck, and vice versa.

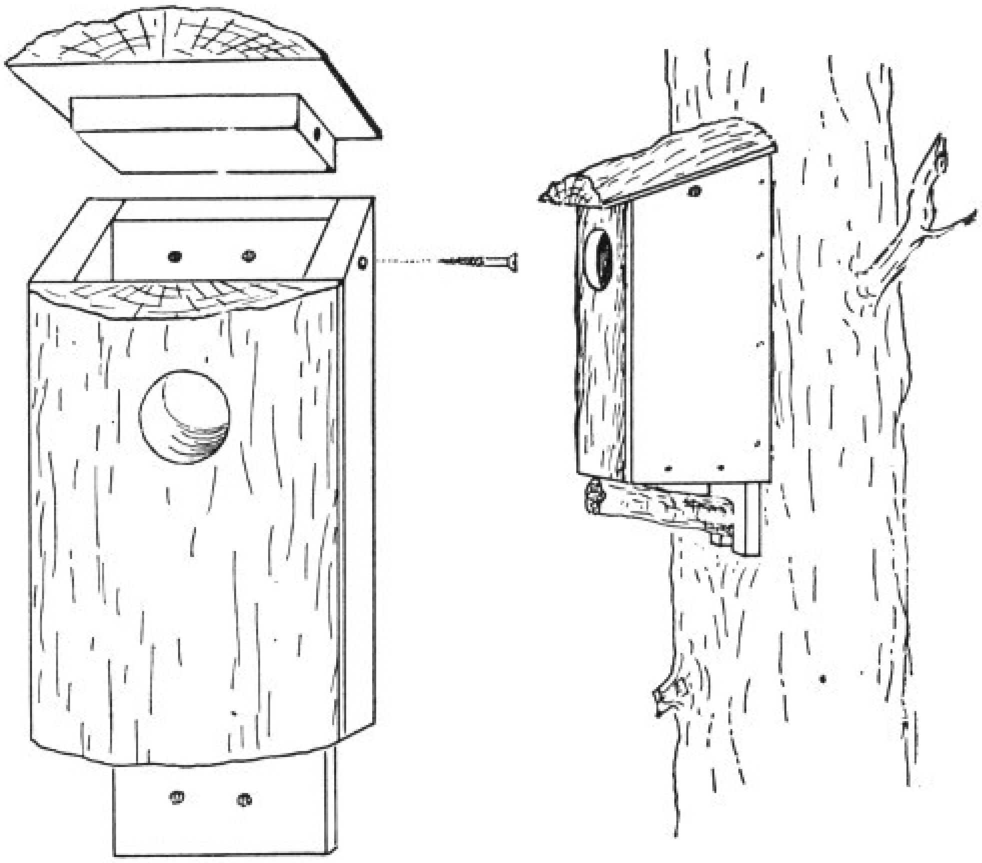


FIGURE 3. An easy-to-make box, especially suitable for ducks. Front and top are slab wood, the rest weathered boards. A close-fitting cleat screwed to under side of top, as shown, keeps top in place with help of one easily removable 2½-inch screw. See Table I, page 16, for dimensions.

The location is of first importance. This should be a secluded wooded stream or body of water. The stump or tree which is to form the support for the house, and also the entrance to the house itself, should be in plain sight from the water. It may be a hundred feet from the nearest shore, but the nearer the shore the better. A lone trunk, or one of several on the edge of a wood, will do. Avoid placing the house too near the ground. Further specifications are given in Table I and Plate V. 28

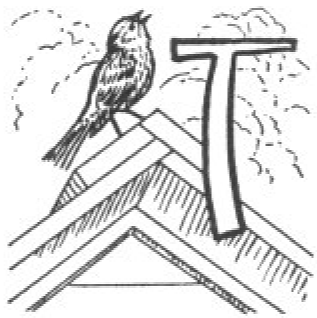
Hawks and Owls

We have alluded to undesirable tenants, meaning usually, or in particular, the English Sparrow and European Starling. There is also another class of possible bird-house occupants to be dealt with—the owls and the Sparrow Hawk. The owls and the hawks are usually considered taboo on account of their fondness for the smaller birds which most persons wish to encourage. One does not ordinarily keep cats and canaries in the same restricted and common area and expect pleasing results. Yet, it may be quite another thing if an estate is large enough to provide sufficient wild cover. Owls and hawks are as interesting as other birds, and a wood, suitable in size and character, which lacks the quota of owls or hawks natural to it will always be lacking in one of its most proper assets and characteristics. To the true nature lover a great wild forest from which owls are excluded might seem, at best, an arboretum where there might as well be a name tag on each tree-trunk.

An occasional nesting site for owls in a wood of many acres, an appropriate box or two for Sparrow Hawks in a waste tract or along a few hundred yards of woodside will invite us to visit these places more often and will provide a new zest to the visits. At the same time, we are but following nature’s way of balancing the wildlife. Nor will the smaller birds be seriously affected; there may be a tendency for many of them to move in a little closer toward our dwellings for increased safety.

A house for small owls and the Sparrow Hawk should follow the lines of the house illustrated in Plate V, size of entrance and other dimensions being given in Table I.

The Common House Finch



The several western and extreme southwestern forms of the native House Finch may, for our present purposes, be grouped with birds like the Robin and Phoebe which find such a number and variety of chance but suitable nesting sites that to provide still others may seem superfluous effort. And yet, to see the bright red of the Common House Finch and to hear his cheery song, say, in the heart of a city like Denver where one looks only for English Sparrows, is to be tempted to offer this citizen a more “desirable property” than the water-spout or other chance nook or cranny in which he may otherwise elect to build. The most successful is the open or semi-open type of nesting box, as shown in Plate VI, Nos. 1 and 2.

Robin and Phoebe

These birds are not classed as bird-house tenants. When they nest in a building, it is nearly always a deserted human dwelling or some other structure made originally for man’s own use. In other words, the only sort of bird house at all likely to attract Robin or Phoebe would be one of cavern-like proportions in keeping with one type of natural site which both species favor, especially the Phoebe.

The architecture of most human dwellings is such that either Robin or Phoebe would find nesting-sites, as they often do, over windows, under porches, or about eaves. But birds are not very considerate of the human liking for cleanliness, and their nests therefore are often placed where we least desire them. To lessen that chance and to furnish nesting-sites when they do not otherwise occur on a given dwelling or outbuilding, the following suggestions for nesting shelves are offered. The idea of nesting shelves is not a new one, and experience shows that an effective nesting shelf may be of almost any description.

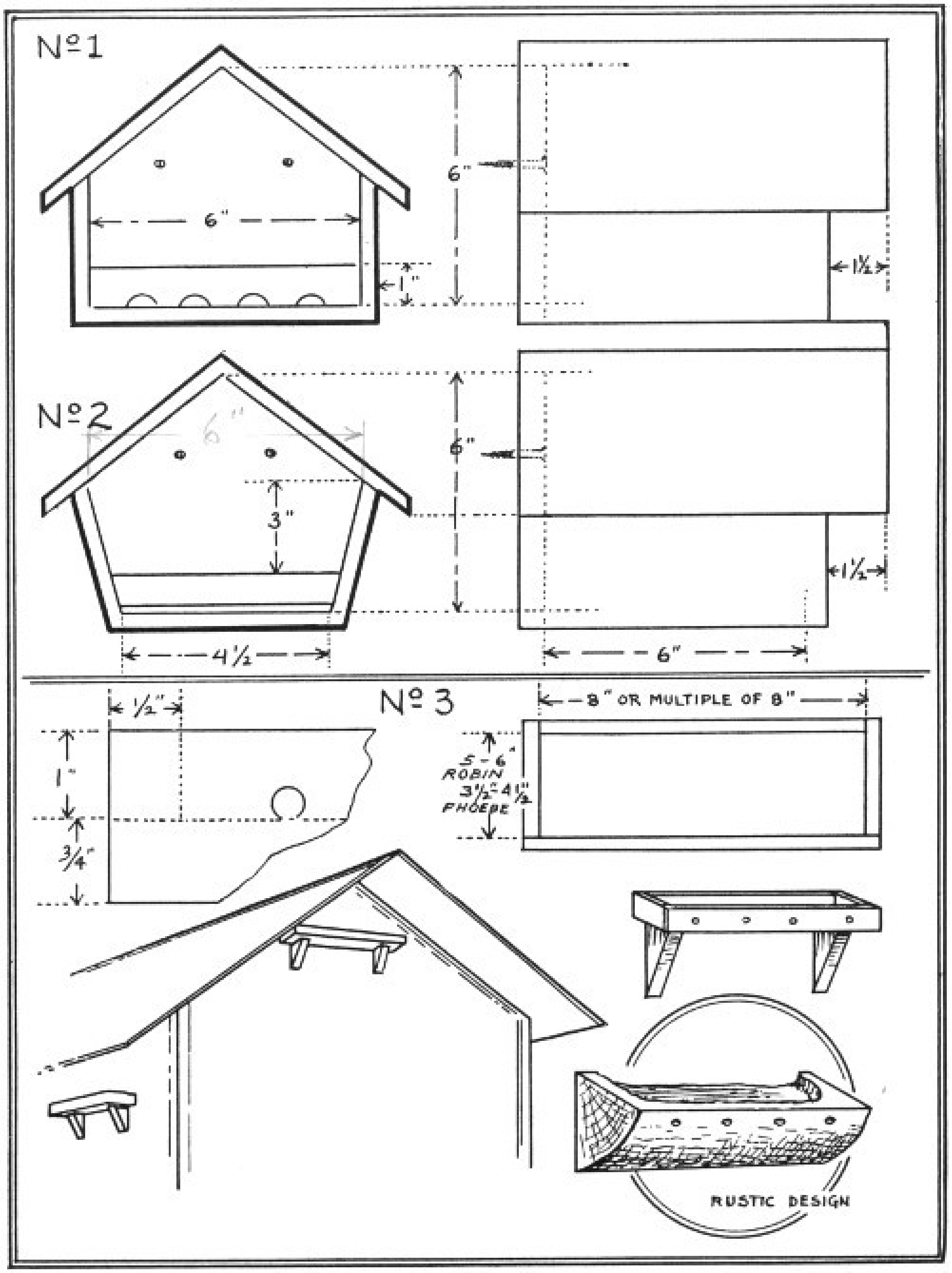


PLATE VI. For the House Finch, Robin, and Phoebe

1 and 2, Nest box for House Finch. The front is left entirely open, except for a cleat to hold the nest in place. Drainage holes may be bored in the bottom of the cleat, or the cleat may be raised a quarter of an inch above the floor level. 3, Nesting shelves for Robin and Phoebe. See text for directions. The raised rim shown here is not always essential, but it may aid the

birds to get a start, since high winds are inclined to blow away the first straws from exposed sites.

The accompanying illustrations will fully explain themselves. 31
However, it is well to emphasize here that the simplest possible shelf, if only a mere cleat, is all that is really required—5 to 6 inches wide for the Robin, 3½ to 4½ inches wide for the Phoebe; any length of 8 inches or over for either bird.

The Robin often nests more than 25 feet up, the Phoebe seldom so high. Place the Phoebe's shelf 8 to 20 feet up, the Robin's 8 to 30 feet. Should the locality be in the country, one must chance a Phoebe claiming a shelf intended for Robins; either species should prove desirable as a tenant. Certainly cultivators of cherries, currants, and other small fruits may well console themselves should Phoebes become their uninvited guests! The Phoebe is a true flycatcher and has none of the Robin's special fondness for garden fruits.

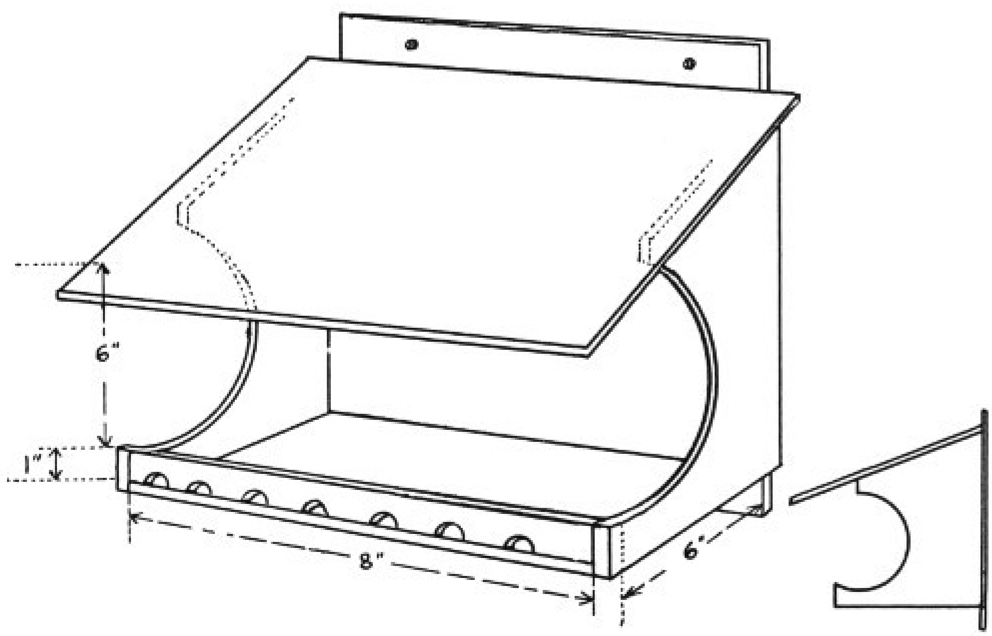


FIGURE 4. A roofed nesting-shelf for Robins and Phoebes. Length may be 8, or any multiple of 8, inches. Several ¼- or ⅜-inch holes may be bored through the floor for drainage, instead of in the rim as shown here.

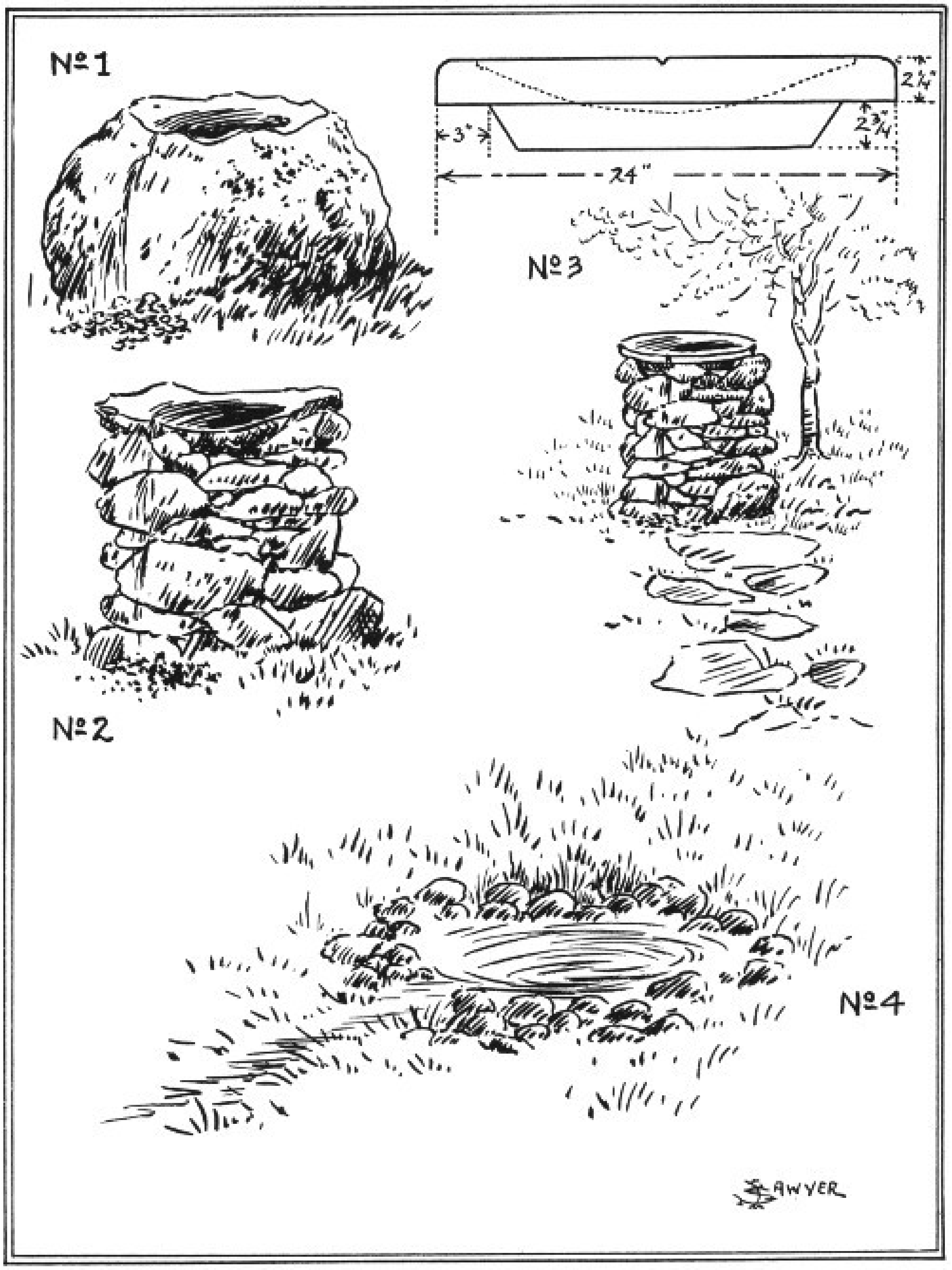
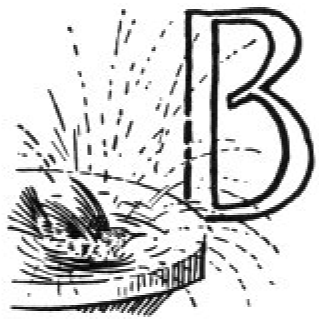


PLATE VII. Bird Baths

1, Natural boulder bath; 2, natural flat stone bath; 3, cement bath on a stone foundation; 4, a ground-pool bath of cement with a border of small boulders.

Bird Baths



Birds are inveterate bathers. Bathing is the daily habit of Robins, Catbirds, goldfinches, Song Sparrows, and most other small species, whenever facilities are available. Artificial baths are most important where other bathing places are distant or inadequate. In times of drought, birds will resort so eagerly to baths as to form an almost continuous daily procession.

The bird bath lends itself to endless variations in size, shape, style, material, and cost. Often one may find a large boulder which, at the expense of moving to the desired spot, will prove a ready-made bath if it has a shallowly concave side. Or such a water basin may be chipped out of a rock by a stone mason. Smaller stones, flattened and more or less scooped, are common along many streams. One of these stone slabs, mounted on a pile of supporting stones, makes an excellent bath. Failing that or as a matter of taste, a massive shallow basin may be cast in cement to take the place of the natural slab. A pool may be provided by lining with cement and surrounding with stones a prepared spot in lawn or garden. See illustrations. If desired, running water may be piped to any style of bath. Whatever type the bath may be, the following rules strictly apply.

1. Depth of water should be graduated from nothing at the edge of bath or pool to not more than 2½ inches at its deepest; except that in the case of the larger ground pools it may be graduated up to 5 inches.

2. The bath must be swept or sponged out daily or as often as it becomes much befouled.

3. Inside of bath should be rough to allow the birds a sure foothold.

4. If the bath is on or near the ground, no shrubbery or other possible concealment for cats should be within 25 feet of it. It is well also to have a branchy tree within a few yards of the bath or pool, so that the bathers when alarmed, may easily reach a place of safety, for their wet plumage will prove a handicap in longer flights.

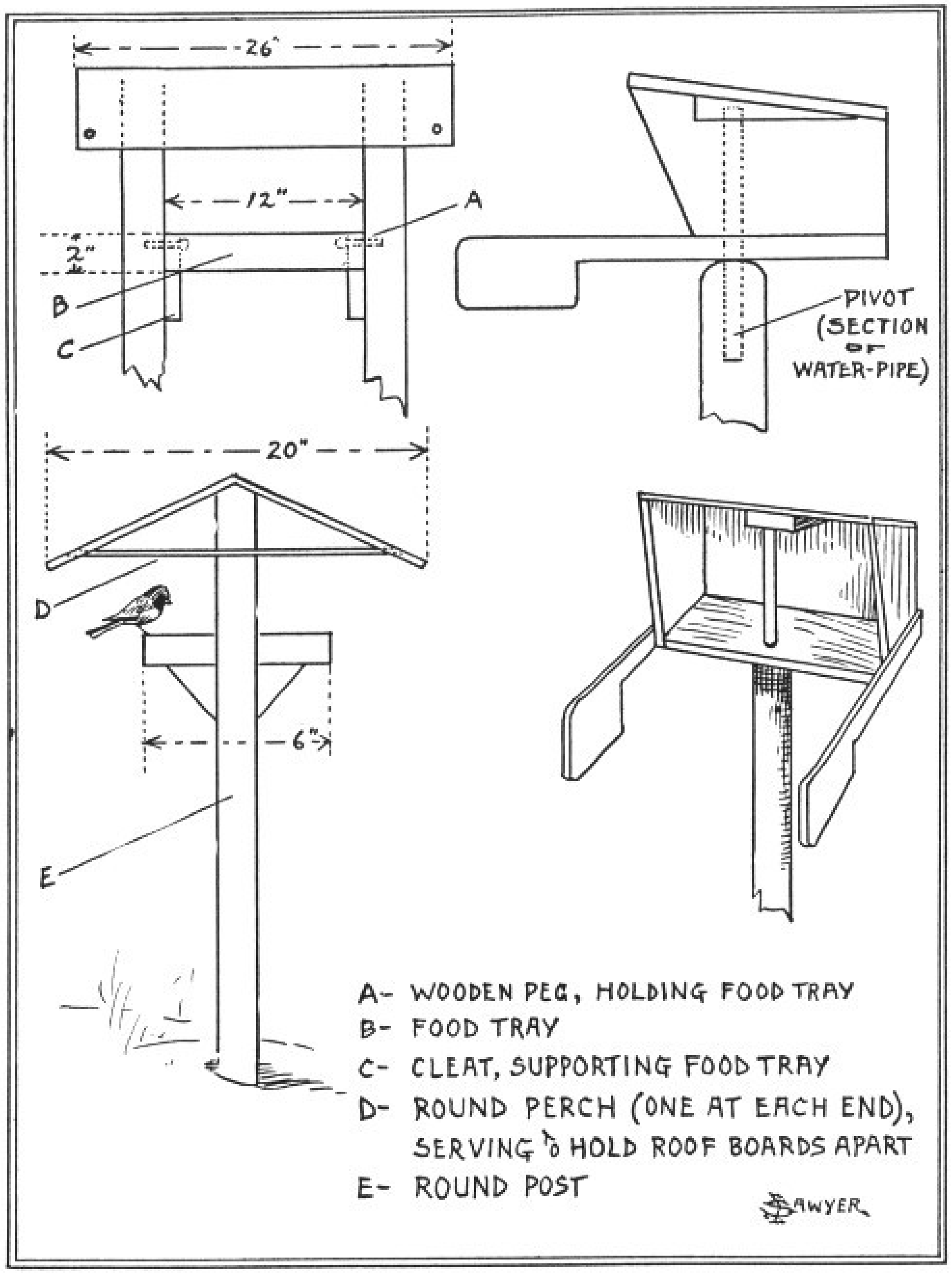


PLATE VIII. Types of Feeding Stations

Trays may be partitioned for different foods. Suet is sometimes tied to the supporting post. The swinging station, shown at the right, always faces away from the wind, but must be carefully balanced at the point of pivoting.

A—WOODEN PEG, HOLDING FOOD TRAY
PIVOT (SECTION OF WATER-PIPE)
B—FOOD TRAY
C—CLEAT, SUPPORTING FOOD TRAY
D—ROUND PERCH (ONE AT EACH END), SERVING TO HOLD ROOF BOARDS APART
E—ROUND POST

Food Stations

Food is the chief problem of winter birds. Cold alone is scarcely a menace, while snow and sleet are chiefly harmful only when they cover up the food. Given proper food, the only real requirement for a feeding station from the birds' point of view is that it shall keep the food available, as by providing a roof to shed snow and ice. Among the wide variety of birds which frequently patronize food stations, various members of the sparrow and finch family, which includes the grosbeaks, juncos, and crossbills, vie with nuthatches, chickadees, woodpeckers, and blue jays as the most dependable boarders.

For winter birds in the northern states, it is well to have the station in place and stocked with food as early as the first of November. These birds are then beginning to establish hunting grounds and routes, from which they will not stray all winter. Earlier in the fall, as again in spring, ground feeding is the better method. In this, scatter the food (millet, hemp seed, and so on) in the back yard, along the fence line or at the edge of shrubbery or thicket. Juncos, Towhees, Song, Fox, and Tree Sparrows, and many others will benefit. Eagerly devoured by the waxwings are dried currants and dried raisins. Nearly all birds are fond of suet. Tie sizable chunks of suet to trees and to posts of food stations; this appeals especially to woodpeckers, nuthatches, and chickadees. Other standard foods are millet, hemp seed, sunflower seed, cracked corn, and bread crumbs. In addition, chaff and oats may be scattered on the ground for quail, grouse, pheasants, Horned Larks, Snow Buntings, longspurs, and others in localities wild enough for these birds. Such feeding is particularly desirable when the snow is covered by an icy crust. The food may be scattered under brush shelters, made of branchy tree limbs loosely and irregularly stacked and roughly thatched with pine, fir, spruce, or other conifer to keep out excess snow.

The care of a food station consists mostly in keeping up a supply of the proper foods and cleaning out the food trays as often as the condition suggests. A small separate tray of coarse sand will provide the grit many birds require. A hopper arrangement for feeding grains aids in keeping the food supply clean and it helps prevent the scattering of seeds.

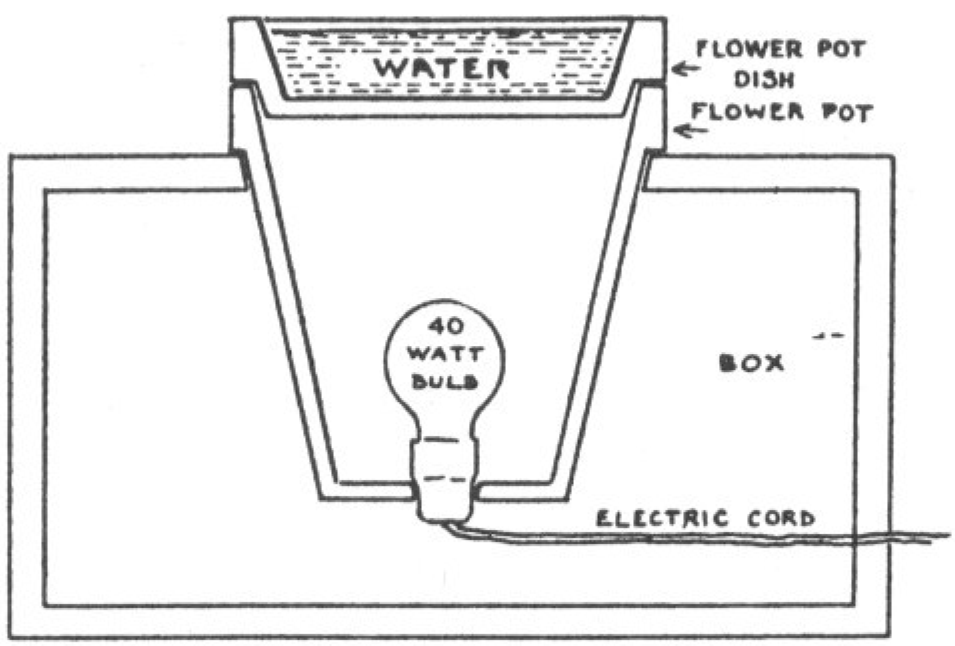


FIGURE 5. Cross-section of a drinking and bathing station for winter use. A flower pot of 8-inch diameter at the top is recommended. If a much larger pot is to be used, a more powerful light bulb may be required to keep the water free of ice in sub-zero weather. If the water is found to be overheated, its temperature may be reduced by placing wedges between the rims of the flower pot and dish or between the rims of flower pot and box.

Wood Ducks

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- Corrected a few palpable typos.
- Included a transcription of the text within some images.
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